DRILL 26

NFPA 1001 Level 2 (JPR 5.3.3) / NFPA 58

ADVANCED FIRE SUPPRESSION

FLAMMABLE GAS / PROPANE TANK COOLING & VALVE SHUTOFF

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear, SCBA (On Air) & Flash Hood

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2 Handline Attack Teams (4–6 Firefighters)

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to safely manage a pressurized **LPG / Propane Tank** fire or leak, recognize signs of an impending **BLEVE (Boiling Liquid Expanding Vapor Explosion)**, advance toward the tank using **dual interlocking wide-fog streams** for thermal shielding, cool the vapor space of the cylinder, and execute a manual shutoff of the main control valve. Propane tanks (100-lb cylinders, 500-to-1,000 gallon domestic "pigs") are universal heating fuel sources across rural New Brunswick.

📦 TOOLS & APPARATUS SETUP

- Pumper supplying two 1.75" (45mm) attack lines at 100 PSI nozzle pressure.
- 2 combination fog nozzles set to wide-fog shield pattern (minimum 30–60 degrees).
- Training propane tank prop (20-lb BBQ tank, 100-lb cylinder, or scrap 500-gallon tank simulator).
- Water supply established (minimum 1,500+ gallons on site before starting live water evolutions).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **THE BLEVE DANGER:** If flame impinges directly on the **vapor space** (top 20–50% of the tank) without cooling water, the steel weakens and can violently rupture within **8 to 15 minutes**.

⚠️ **BLEVE Warning Signs:** A rising pitch in the sound of roaring gas venting from the relief valve, or visible discoloration/bulging of the tank shell. **EVACUATE IMMEDIATELY (Minimum 500 metres / 1,500 feet)!**

⚠️ **Continuous Water Supply Mandatory:** Never approach a burning pressurized vessel unless an uninterrupted, positive water supply is guaranteed.

🛑 **THE STOP RULE: Shout "STOP!" immediately if either hose line loses water pressure, if nozzles fall out of interlock, or if a firefighter trips during the approach.**

PHASE 1: PROPANE PROPERTIES & APPROACH STRATEGY (15 MINUTES) — SLOW DRY RUN

1. **Properties of Propane (LPG):**
   - Vapor density is **1.5 (heavier than air)**; leaking gas pools in low ditches, basements, and around foundations.
   - Expansion ratio: **270 to 1** (1 liter of liquid expands into 270 liters of explosive gas!).
2. **The Dual-Line Interlocking Shield:**
   - Two 1.75" lines advance side-by-side.
   - Both nozzles set to **wide-fog pattern (45–60 degrees)**.
   - The two fog streams **overlap by 1 to 2 feet in the center**, creating a solid, impenetrable thermal water wall protecting the firefighters behind them.
   - The fog stream pushes radiant heat and flames away while entraining fresh air for the crew.

DRILL 26: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Advanced Fire Suppression • Practical Application

PHASE 2: COORDINATED APPROACH WALKTHROUGH (15 MINUTES) — BUDDY COACHING

1. Both nozzle teams kneel or crouch side-by-side, shoulder-to-shoulder.
2. Team Leader (between the two nozzlemen) coordinates cadence: *"Step... Step... Step..."*
3. Move forward at a slow, deliberate pace. Nozzlemen keep streams focused on the **upper shell (vapor space)** of the propane tank to keep the steel cold.
4. If the relief valve is burning, **DO NOT extinguish the flame** until the valve can be shut! Extinguishing the flame before shutting the leak creates an invisible cloud of heavier-than-air gas that will ignite into a catastrophic vapor cloud explosion.

PHASE 3: VALVE ISOLATION & SAFE RETREAT (20 MINUTES) — TIMED / STRESS

1. As the crew reaches arm's length from the tank:
   - Both nozzlemen open their fog patterns wide to deflect flames around the sides.
   - The Team Leader (or backup firefighter) reaches between the streams with gloved hands.
   - Grasp the handwheel of the service valve and **rotate clockwise until fully closed**.
2. **Confirm Extinguishment:**
   - Once the valve is closed, the flame will starve and extinguish.
   - Continue cooling the hot tank shell with fog streams until steam generation stops.
3. **The Coordinated Retreat:**
   - **NEVER TURN YOUR BACK ON THE TANK!**
   - Keeping both fog streams flowing toward the tank, the crew backs up step-by-step in unison until clear of the hot zone (minimum 50 feet away).
   - Nozzles are shut down slowly to prevent water hammer.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Turning off nozzles when the fire goes out:* If the valve didn't seal 100%, gas continues leaking; shutting off the water shield leaves you unprotected against re-ignition. Keep the shield up while backing out!
- ❌ *Splitting the fog patterns:* If nozzlemen pull apart, a gap opens in the center of the water shield, allowing superheated flames to blast between them. Keep streams firmly interlocked.
- ❌ *Approaching from the ends of the tank:* If a cylinder ruptures, the convex end caps shoot outward like missile rockets. **Always approach from the sides (perpendicular to the tank ends)!**

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How loud was the water shield roaring during the approach?", "Why must we approach the sides of the tank rather than the ends?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hoses drained, dried, and repacked.
- ☐ Nozzles flushed and set to department default pattern.
- ☐ SCBA facepieces disinfected; cylinders refilled.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS